How We Tested

We segregated the motherboards into two distinct platforms—Intel and AMD. Based on the price, we split each into three categories: low, mid and high.

The low-end category consisted of motherboards that fall within the Rs 5,000 range, whereas the mid-end encompassed those with a price tag of up to Rs 10,000. Anything above this was slotted in the high-end category.

All the low-end boards were benchmarked using 512 MB DDR 266 MHz, a Pentium 3.06 GHz or an Athlon XP 2400+ processor and onboard graphics.

The mid-range boards were tested using 512 MB of DDR 400 MHz, a Pentium 3.2 GHz or an Athlon XP 3200+ CPU and a Gigabyte Radeon 9600 Pro graphics card.

A Pentium 3.2 GHz or Athlon XP 3200+ CPU, 512 MB of DDR 400 and a Gainward FX 5950 Ultra display card formed the test bed for the high-end category.

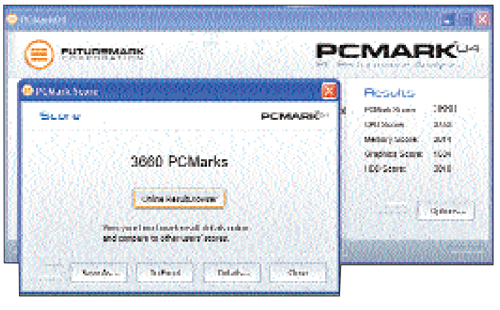
All the boards were tested using Seagate's 40 GB 7200-rpm IDE drives. We chose Windows XP Professional with SP-1 as the operating system, and installed the latest drivers for chipsets and displays.

Benchmarks

We segregated these tests into real-world and synthetic tests. Here, many new benchmarks such as PCMark 2004, Ziff Davis Suite 2004 and SiSoft Sandra 2004 were included.

Synthetic tests

PCMark 2004: A new avatar of the older PCMark 2002, this test has 70 new tests that benchmark the entire system and give a unified score. You can also run tests that give individual sub-system



PCMark 2004 is a complete system benchmarking suite

scores for CPU, graphics, memory, hard drives, etc. The tests mainly comprise applications such as WinZip used on a day-to-day basis. Scores are generated depending on the system performance in each individual tests.

3D Mark 03 and 3D Mark 2001 SE: Both these synthetic benchmarks from Future Mark are widely used to benchmark graphical sub-systems. Both were patched with the latest upgrade patches from FutureMark.

3D Mark 2001 SE was not used on high-end systems, just as 3D Mark 2003 was not run on low-end motherboards since it would skip most of the tests on onboard graphics—both were run on mid-range systems.

SiSoft Sandra 2004: This is the same old benchmarking tool with newer features and added tests. The interface has also been touched up a little. There's a new user interface and an extra benchmarking tool for USB drives, cache memory, etc. The internal database has been revised for comparing with new processors,

SiSoft Sandra 2004 has a new user interface and additional tests bundled along

memory standards and newer chipsets. We logged scores particularly related to the CPU, memory and the disk drive.

Real-world tests

Ziff Davis Suite 2004: The Ziff Davis Suite 2004 consists of the Multimedia Content Creation and Business Winstone benchmarks. While PCMark 2004 just uses the necessary sections of Adobe Photoshop, MS Office, etc, this test suite installs entire applications to simulate real world conditions. Multimedia Content Creation runs Adobe Photoshop, Adobe Premier, sound-editing applications, etc; whereas Business Winstone runs MS Office, and makes use of features such as file-zipping, etc, to simulate its performance when used on a daily basis. The unified score reflects the performance of the system as a whole unit.

UT 2003: Based on DirectX 8.1, it was used to stress the graphic sub-system of the motherboard. Here, we logged the Flyby demo fps. *UT 2003* was used only for the mid-range and high-end category. This benchmark is quite heavy in terms of graphical requirements, hence, running it on low-end motherboards would not yield proper results.

Serious Sam 2: *Serious Sam 2* was used to check how well the graphical sub-system performs in OpenGL environments. We logged a score for resolutions up to 1024 x 768 x 16 for low-end motherboards that used onboard graphics. We tested the high-end motherboards for a resolution of 1600 X 1200 X 32.
The logic used was that the enthusiast would definitely like to play games at higher resolutions on the mid-range and high-end boards.

Quake III Arena: This is the grand daddy of all game benchmarking tools. It's based on the OpenGL environment, and holds fort due to its super-oiled engine that scales true to any hardware changes done to the system. We followed the same procedure used in *Serious Sam 2* to log scores for *Quake III Arena* too.

The gaming scores mentioned in the table are the averages of frames per second (fps) obtained at various resolutions.

Video Encoding: We encoded a 50 MB MPEG file to DIVX using VirtualDub and used the DivX 4.01 codec for the process. The time taken to complete the job was logged, and then compared.